

PDA College Of Engineering
Department Of Computer Science and Engineering

SYNOPSIS OF THE PROPOSED MAJORPROJECT (22CS65)

ON

**InstaGlide: Seamless Customer Interaction & Automated Messaging
SaaS for Instagram**

Submitted by:

VTU No.	NAME	SEMESTER	SECTION
3PD23CS065	Md Ahmed Raza Sofi	V	A
3PD23CS050	Hisham Khuram	VI	A
3PD23CS070	Mohammed Abdul Maaz Hanan	VI	A

Under the Guidance of

Guide Name: Dr. Anita Harsoor

Affiliation: CSE Department, PDA College of Engineering

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Department Vision Mission Statement

Vision

To become a premier department in Computer education, research, and to prepare highly competent IT professionals to serve industry and society at local and global levels.

Mission

- To impart high quality professional education to become a leader in Computer Science and Engineering.
- To achieve excellence in research for contributing to the development of the society.
- To inculcate professional and ethical behaviour to serve the industry.

ABSTRACT

"InstaGlide" is a comprehensive Software-as-a-Service (SaaS) application meticulously designed to automate and optimize customer interactions on Instagram. Built specifically for the rapidly growing creator economy and modern digital businesses, the application serves as a dynamic bridge between content and client acquisition. It utilizes the Meta/Instagram Graph API to actively listen for specific, user-defined keywords within post comments and Direct Messages (DMs), using these keywords as triggers to fire off customized automation sequences.

What distinctly sets InstaGlide apart from traditional automation platforms is its native integration of Large Language Models via the OpenAI API, branded within the platform as "Smart AI". Instead of relying exclusively on rigid, pre-programmed, single-message responses, InstaGlide empowers users to deploy an intelligent virtual agent that can hold dynamic, multi-turn conversations. This Smart AI tracks the customer's chat history, understands context, and acts effectively as a digital "DM Setter" or sales closer, nurturing leads toward a specific goal based on a custom prompt provided by the user.

Under the hood, the system is engineered using a cutting-edge, full-stack JavaScript/TypeScript architecture. It is powered by Next.js 14 utilizing the App Router, providing a blazing-fast, server-rendered frontend. The backend relies on an event-driven architecture utilizing real-time Webhooks to capture Meta's payload data instantly. Data persistence is handled securely via a serverless Neon PostgreSQL database managed by the Prisma ORM, while user authentication and subscription billing are seamlessly handled by Clerk and Stripe, respectively.

INTRODUCTION

In the modern digital landscape, interacting with customers on social media platforms like Instagram is a fundamental requirement for business growth and brand loyalty. For content creators, influencers, and digital storefronts, the comment section and Direct Messages (DMs) serve as the primary funnels for lead generation and client acquisition. However, a significant operational bottleneck occurs as a business scales: maintaining a high level of real-time, one-on-one interactivity with a growing audience quickly becomes physically impossible. To bridge this gap, businesses are often forced to manually reply to thousands of messages or hire expensive human sales representatives, commonly known as "DM setters," to manage their inboxes.

While chatbot automation platforms emerged around 2015 (with industry leaders like ManyChat) to solve this exact problem, the current market offerings suffer from critical limitations. Traditional bots rely heavily on rigid "If-This-Then-That" logic, sending static, pre-written text blocks that feel highly robotic to the end consumer. They lack contextual awareness and struggle to handle nuanced follow-up questions from potential leads. Furthermore, these existing legacy platforms utilize restrictive, tiered pricing models that penalize a business for growing; for instance, managing a list of one million contacts can cost a business upwards of \$5,000 every single month, making it a highly unreasonable expense for independent creators.

The "InstaGlide" project is conceptualized as a next-generation, highly scalable alternative designed to disrupt this space. By combining modern web development frameworks with generative artificial intelligence, InstaGlide offers a much smarter way to automate Instagram interactions. At its core, InstaGlide provides a visual automation builder where users can attach keyword "triggers" to specific Instagram media assets or direct messages. When a potential lead interacts using that specific keyword, the platform instantly registers the event via webhooks and deploys a tailored action, referred to as a "listener".

The true innovation of InstaGlide lies in its "Smart AI" capability. Instead of deploying a static response, businesses can configure the AI by providing a system prompt (e.g., "You are a professional, engaging DM setter for a fitness business"). The system seamlessly tracks the conversation history, allowing the AI to converse naturally, answer specific product questions, and intuitively guide the user toward a specific conversion or external link. Coupled with a flat-rate subscription model that offers unlimited automations and contacts, InstaGlide represents a highly optimized, cost-effective, and intelligent solution for the future of social media marketing.

PROBLEM STATEMENT

In the rapidly expanding digital landscape, interacting with customers through Instagram Direct Messages (DMs) and post comments is a critical driver for business growth and lead generation. However, businesses and content creators face several major roadblocks when attempting to scale these interactions:

- **The Scalability Bottleneck:** As an audience grows, manually tracking, reading, and responding to an influx of DMs and comments becomes physically impossible to manage. To maintain a high level of interactivity, businesses are often forced to rely on manual human labor, such as hiring expensive sales representatives or "DM Setters" to monitor inboxes.
- **Prohibitive Costs of Existing Automation:** While legacy chatbot automation platforms (such as ManyChat) exist to solve this communication gap, they utilize restrictive, tiered pricing models. As a business acquires more users and its contact list grows, the operational costs skyrocket. For instance, managing a list of one million contacts on traditional platforms can cost a business up to \$5,000 every single month, making it an unreasonable expense for many independent creators.
- **Rigid and Static Interactions:** Current automation tools rely heavily on basic keyword triggers that simply deploy a single, pre-written message. These traditional bots lack contextual awareness and cannot hold dynamic, multi-turn conversations. Because they act rigidly, they fail to effectively replace human closers when potential leads ask nuanced follow-up questions.

EXISTING AND PROPOSED SOLUTIONS

Existing Solution: Businesses currently rely on manual human labor or static chatbot builders (like ManyChat). These bots trigger basic "If-This-Then-That" static text replies, which feel robotic and fail to adapt to complex customer follow-up questions.

Proposed Solution: The proposed solution, **InstaGlide**, is a next-generation Software-as-a-Service (SaaS) application explicitly engineered to capture and nurture leads on Instagram through intelligent, automated sequences. Unlike legacy competitors such as ManyChat, which penalize scale by charging exorbitant tiered fees (e.g., \$5,000 per month for a million contacts), InstaGlide introduces a disruptive, flat-rate pricing model. Users are empowered to run unlimited automations and manage unlimited contacts without incurring escalating costs, making high-volume lead generation accessible and predictable for creators and businesses.

At the foundation of InstaGlide is a blazing-fast, visual automation builder utilizing **Optimistic UI** to provide an instantaneous, seamless user experience. The system is built on a straightforward "When -> Then" logic flow. Users define a **"Trigger"** (the *When*), which actively listens for specific, user-defined keywords. These triggers can be attached to two primary events: a user leaving a comment on a specific post, or a user sending a direct message (DM) containing the keyword. Once the trigger condition is met, it fires a corresponding **"Listener"** or action (the *Then*), which dispatches the automated response back to the potential lead.

The true core differentiator of the InstaGlide platform is the native integration of Large Language Models (LLMs) via the **"Smart AI" feature**. Rather than limiting businesses to single, rigid text responses, InstaGlide users can hand the conversation over to a custom-trained AI agent powered by OpenAI. Users provide a system prompt (e.g., "You are a professional DM setter for a fitness business"), instructing the bot on its exact persona and end goal. The Smart AI then effectively replaces human "DM Setters" by tracking conversation history, dynamically answering follow-up questions, and organically guiding the customer toward a specific call-to-action or link. Furthermore, the system is architected with specific API workarounds ensuring these AI responses land directly in the customer's **primary inbox**, rather than being hidden in the obscure "requests" tab.

METHODOLOGY

The architecture of InstaGlide relies heavily on an **event-driven backend** powered by the Instagram Graph API and real-time Webhooks. The application's core infrastructure is built using Next.js 14 server actions, a serverless Neon PostgreSQL database, and the Prisma ORM for structured data modeling. When a user connects their professional Instagram account via an OAuth flow, the system securely provisions and stores long-lived access tokens to authenticate future API calls. From there, the application subscribes to Meta's webhooks, specifically listening for changes in messages and comments.

The execution flow begins the moment an interaction occurs on Instagram. Meta's API instantly dispatches a JSON payload to InstaGlide's designated webhook endpoint. The system first parses this payload through a **Keyword Matcher** algorithm. The algorithm scans the text of the incoming comment or DM in a case-insensitive manner to check if it matches any active automation keywords stored in the PostgreSQL database. If no match is found for a new interaction, the process silently terminates; however, if a keyword is matched, the system identifies the specific automation sequence associated with that trigger.

Once the sequence is identified, the backend logic splits into two distinct processing pathways based on the **Trigger Type**.

- **For Comment Triggers:** The system verifies if the exact media asset (the specific Instagram post) is registered to the automation. Upon validation, it utilizes a "Private Replies" API workaround to securely send a direct message to the commenter using their specific `comment_id`.
- **For Direct Message Triggers:** The system bypasses post-validation and immediately evaluates the requested action, termed the **Listener**.

The final phase of the methodology involves dispatching the **Listener Actions**. If the user configured a standard message, the system fires an Axios POST request via the Instagram API containing the static prompt. If the user selected **Smart AI**, the methodology executes a complex, state-aware loop. The backend queries the database to determine if the interacting customer is a "new lead" or a "returning lead". For a returning lead, the system actively retrieves prior chat history to establish context. This historical data, paired with the creator's custom training prompt, is then fed into the OpenAI API (`chat.completions.create`). The LLM generates a personalized response (capped at two sentences to maintain natural pacing), sends it to the user's primary DM inbox, and logs the new interaction back into the database to preserve conversational memory for future replies.

SYSTEM REQUIREMENTS

To successfully develop, test, and deploy the InstaGlide application, a specific set of hardware, runtime environments, and third-party developer accounts are strictly required.

Development Environment & Runtime

- **Runtime Environment:** The project is optimized to run on **Bun**, a fast JavaScript all-in-one toolkit, rather than standard Node.js/npm.
- **Development Tools:** A modern IDE (like Visual Studio Code) and a local tunneling service like **Ngrok**. Ngrok is explicitly required during the development phase to expose the local development server (localhost) to the internet, allowing Meta's live webhooks to reach the local API endpoints.

Third-Party Services & API Accounts

- **Meta Developers Account:** A registered developer account with Meta is mandatory to create a "Business" app. This account must be linked to a Facebook Page and an active Instagram Business account. It is required to access the Instagram Graph API, configure webhook callback URLs, and set up the OAuth flow for users to connect their accounts.
- **OpenAI Account:** Required to generate the conversational responses for the "Smart AI" feature. Developers must set up an active billing account with a minimum budget (e.g., \$5) to access the API keys required for the chat.completions endpoint.
- **Neon Database Account:** Required to provision a serverless PostgreSQL database to securely store user, automation, and conversational data.
- **Clerk Account:** Required for managing user authentication, security, and rendering sign-in/sign-up components seamlessly.
- **Stripe Account:** Required to configure the payment gateway. Developers must create a subscription product (e.g., "Pro Plan") and extract the Price ID and Secret Keys to handle SaaS billing.
- **Vercel Account:** Required for the final production deployment and hosting of the Next.js application.

SYSTEM SPECIFICATIONS

The system is architected as a modern, full-stack web application utilizing a cutting-edge JavaScript/TypeScript ecosystem.

Frontend Architecture

- **Framework:** The user interface is built on **Next.js 14.2.2**, heavily utilizing the **App Router** for structural routing and server-side rendering.
- **Styling & UI Components:** The application uses **Tailwind CSS** for utility-first styling and **Shadcn UI** for highly customizable, accessible, and pre-built interactive components (such as sheets, popovers, and accordions). Custom SVG icons and gradients are widely used to maintain a premium SaaS aesthetic.
- **State Management & Caching:** The frontend employs a sophisticated dual-state management system. **Redux Toolkit** is used as the global state manager specifically to handle the complex state of the visual automation builder (managing triggers and listeners). Concurrently, **TanStack React Query** is heavily utilized for data fetching, caching, and executing complex **Optimistic UI** updates. This ensures the interface feels instantaneous, updating visually before the database confirms the mutation.
- **Form Handling:** Complex user inputs, specifically for customizing the AI prompts and messaging, are handled using react-hook-form paired with **Zod** for strict schema validation and type safety.

Backend Architecture

- **Server Actions:** Instead of relying entirely on traditional REST APIs, the application predominantly uses Next.js **Server Actions** to handle backend business logic, database queries, and secure data mutations directly from client components.
- **Webhook Endpoints:** Standard API routes are reserved specifically for listening to external events. A dedicated `/api/webhook/instagram` endpoint handles the payload challenges and real-time message/comment payloads dispatched by Meta. A secondary `/api/payment` endpoint manages the Stripe checkout session redirection.
- **AI Integration:** The backend utilizes the `openai` Node package to send system prompts (customized by the user) alongside the customer's chat history to the AI model, enforcing a strict two-sentence generation limit to mimic natural human texting behavior.

EXPECTED OUTCOME OF THE PROJECT

The expected outcome of the InstaGlide project is the successful deployment of a robust, production-ready SaaS application capable of automating complex Instagram interactions. Upon completion, the platform will offer a comprehensive solution for digital creators and businesses to capture leads without manual intervention.

Key Functional Outcomes:

- **Seamless Onboarding & Monetization:** Users will be able to securely authenticate via Google (powered by Clerk) and upgrade their accounts through a fully functional Stripe checkout portal to access premium AI features.
- **Secure API Integration:** Users will be able to natively link their professional Instagram accounts to the platform. The system will securely request, store, and automatically refresh long-lived access tokens to interact with the Instagram Graph API on their behalf.
- **Dynamic Visual Automation Builder:** The project will yield a highly responsive, optimistic-UI-driven dashboard where users can intuitively build "When-Then" workflows. They will be able to easily map specific keyword triggers to dedicated Instagram posts or direct messages.
- **Intelligent, Autonomous Engagement:** Once an automation is activated, the application will successfully capture real-time webhooks from Meta. It will accurately parse incoming comments or DMs, cross-reference them against the database for keyword matches, and dispatch responses. If the user configures the "Smart AI" listener, the expected outcome is that the platform will not only send an initial message but will effectively track conversation history to hold context-aware, goal-driven conversations, mimicking a human sales representative in the customer's primary inbox.